

RAG Project Concepts Explained with Examples

Big Picture of the Project

Imagine you upload a Biology PDF. The PDF contains information about photosynthesis. When a user asks a question, the system does not directly send the entire PDF to Gemini. Instead, it follows a process: PDF -> Chunking -> Embeddings -> FAISS Search -> Context Retrieval -> Gemini -> Answer.

What is a Chunk?

A chunk is a small piece of a document. Example: PDF Text: Photosynthesis is the process by which plants convert sunlight into energy. Plants use chlorophyll for photosynthesis. Oxygen is released as a byproduct. Chunk 1: Photosynthesis is the process by which plants convert sunlight into energy. Chunk 2: Plants use chlorophyll for photosynthesis. Chunk 3: Oxygen is released as a byproduct. Think of a PDF as a big book and chunks as small paragraphs from that book.

Does Gemini Search the PDF?

No. Gemini never searches the entire PDF. FAISS performs the search. Example: User asks: What is photosynthesis? FAISS searches through hundreds or thousands of chunks and finds the most relevant chunk. Then only those relevant chunks are sent to Gemini. Think: FAISS = Librarian Gemini = Teacher The librarian finds the correct page. The teacher explains the page.

What is Context?

Context is the relevant information retrieved by FAISS. Example: User Question: What is photosynthesis? FAISS finds: Photosynthesis is the process by which plants convert sunlight into energy. This retrieved information becomes the context. Context is not always the final answer. It is the information Gemini uses to create the answer.

How Gemini Generates Answers

Gemini receives: Context: Photosynthesis is the process by which plants convert sunlight into energy. Question: What is photosynthesis? Gemini reads both and generates: Photosynthesis is the process by which plants convert sunlight into energy. The answer is returned through the Gemini API and displayed in Streamlit.

Can Gemini Only Rephrase?

No. Gemini can: 1. Rephrase 2. Merge information 3. Summarize 4. Explain concepts 5. Answer questions Example: Chunk 1: The model used CNN. Chunk 2: The model achieved 95% accuracy. Gemini Answer: The experiment used a CNN model and achieved an accuracy of 95%. Gemini combined information from multiple chunks.

Understanding vs Rephrasing

Suppose the PDF says: The model achieved 95% accuracy. User asks: Was the model successful? The PDF never explicitly says 'successful'. Gemini understands that 95% accuracy indicates good performance and answers: Yes, the model appears to have been successful, achieving an accuracy of 95%. This is understanding, not simple rephrasing.

Why We Need FAISS

Imagine a PDF creates 1000 chunks. Sending all 1000 chunks to Gemini would be: - Expensive - Slow - Inefficient FAISS searches all chunks and returns only the top relevant chunks. Example: Total Chunks: 1000 Retrieved: Chunk 45 Chunk 89 Chunk 112 Gemini receives only those chunks.

Role of Gemini in This Project

Gemini is an intelligent language model. After receiving context from FAISS, Gemini can: - Understand the user's question - Understand the meaning of the retrieved chunks - Combine information - Summarize information - Explain information in simple language - Generate natural-language answers Gemini does not search the PDF. Gemini works on the context provided by FAISS.

Most Important One-Line Summary

FAISS finds the relevant information. Gemini understands the information and explains it.